SG No. 135 D_{4h}^{13} $P4_2/mbc$ [tetragonal]

* plus set: + [0, 0, 0]

Table 1: Wyckoff site: `4a`, site symmetry: `2/m..`

No.	position	mapping
1	[0, 0, 0]	`[1,2,9,10]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[3,4,11,12]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[5,6,13,14]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[7,8,15,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4..`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,11,12]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[3,4,9,10]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,6,15,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[7,8,13,14]`

Table 3: Wyckoff site: `4c`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,9,10]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[3,4,11,12]`
3	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[5,6,13,14]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[7,8,15,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `2.22`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,2,7,8]`
2	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[3,4,5,6]`
3	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,10,15,16]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[11,12,13,14]`

Table 5: Wyckoff site: `8e`, site symmetry: `2..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2]`
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[3,4]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[5,6]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[7,8]`
5	$[0,\ 0,\ -z]$	`[9,10]`
6	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[11,12]`
7	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[13,14]`

continued ...

Table 5

No.	position	mapping
8	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[15,16]`

Table 6: Wyckoff site: `8f`, site symmetry: `2..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,2]`
2	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[3,4]`
3	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[5,6]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[7,8]`
5	$\left[0,\ \frac{1}{2},\ -z\right]$	`[9,10]`
6	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[11,12]`
7	$\left[\frac{1}{2},\ 0,\ z\right]$	`[13,14]`
8	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[15,16]`

Table 7: Wyckoff site: `8g`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[1,7]`
2	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[2,8]`
3	$\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$	`[3,5]`
4	$\left[x+\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	`[4,6]`
5	$\left[-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[9,15]`
6	$\left[x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[10,16]`
7	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	`[11,13]`
8	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$	`[12,14]`

Table 8: Wyckoff site: `8h`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,10]`
2	$[-x,\ -y,\ 0]$	`[2,9]`
3	$\left[-y,\ x,\ \frac{1}{2}\right]$	`[3,12]`
4	$\left[y,\ -x,\ \frac{1}{2}\right]$	`[4,11]`
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ 0\right]$	`[5,14]`
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	`[6,13]`
7	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[7,16]`
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[8,15]`

Table 9: Wyckoff site: `16i`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`

continued ...

Table 9

No.	position	mapping
2	$[-x, -y, z]$	[2]
3	$\left[-y, x, z+\frac{1}{2}\right]$	[3]
4	$\left[y, -x, z+\frac{1}{2}\right]$	[4]
5	$\left[\frac{1}{2}-x, y+\frac{1}{2}, -z\right]$	[5]
6	$\left[x+\frac{1}{2}, \frac{1}{2}-y, -z\right]$	[6]
7	$\left[y+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}-z\right]$	[7]
8	$\left[\frac{1}{2}-y, \frac{1}{2}-x, \frac{1}{2}-z\right]$	[8]
9	$[-x, -y, -z]$	[9]
10	$[x, y, -z]$	[10]
11	$\left[y, -x, \frac{1}{2}-z\right]$	[11]
12	$\left[-y, x, \frac{1}{2}-z\right]$	[12]
13	$\left[x+\frac{1}{2}, \frac{1}{2}-y, z\right]$	[13]
14	$\left[\frac{1}{2}-x, y+\frac{1}{2}, z\right]$	[14]
15	$\left[\frac{1}{2}-y, \frac{1}{2}-x, z+\frac{1}{2}\right]$	[15]
16	$\left[y+\frac{1}{2}, x+\frac{1}{2}, z+\frac{1}{2}\right]$	[16]